

Data Request: Nevada Division of Environmental Protection

Steamboat Hills Geothermal Outflow Monitoring – UNR Thesis Project

2026-09-25

Project background

This request supports a University of Nevada, Reno thesis project studying changes in geothermal outflow at Steamboat Hills, Nevada, following the June 2025 Lower Sinter Terrace eruption. The project maintains a consolidated database and pipeline integrating historic and current chemistry, water levels, and continuous conductivity/temperature monitoring at the Steamboat field, cross-referenced against NDEP's own Steamboat monitoring program and the Temporary UIC Permit UNEV2007204T2025-1 REVISED (June 2025).

What we are requesting

#	Request	Why it matters
1	Daily water-level (or pressure) records for wells 43-33-1, 21-32-1, 21-32-2	These are the deepest, most centrally-located wells named in the UIC permit itself. Currently only 2-3 wells inside the Steamboat field proper have any usable current water-level record at all – this is the single largest blocker to building a real potentiometric surface for the field.

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2	Fumarole inspection logs referenced in the UIC permit	Direct evidence of surface thermal-manifestation change over time, complementing this project's own 2025-2026 spring/fissure field observations.
3	Water levels for observation wells OW-1, OW-2, OW-3 and the stratigraphic test wells (strat 2, 5, 6, 7, 8, 9, 13, 14)	These wells are already registered in our database (from Sorey & Colvard, 1992) but have zero water-level history on file.
4	Current chemistry (major ions, at minimum chloride) for the standing NDEP monitoring network – NDOT, Herz, Boyd, Soccer Field, Eich, Jeppson, Rogers	A current round of chemistry at these already-located wells lets us test whether the historical monitoring signal (chloride, geochemical facies) has changed since the 2025 eruption.
5	More precise surveyed coordinates for any of the above wells, if available	Most well coordinates in our current database come from NBMG/ArcGIS digitization (50-800 m uncertainty), not a real survey.

A specific, time-sensitive ask

We are also planning a real Cl-conductivity calibration sampling campaign at two continuous-conductivity monitoring sites (upstream/downstream on Steamboat Creek) over the coming year. If NDEP has any standing chemistry sampling planned at nearby locations during this window, coordinating sample timing would benefit both programs.

Contact

Please direct any response to [Tyler Irvin / UNR contact information], referencing this request and the Temporary UIC Permit UNEV2007204T2025-1 REVISED for context.